DATABASE NORMALIZATION

Computer Repair Appointment and Service Management System

Normalization Overview

Normalization is the process of organizing the database so that data is stored in appropriate tables, unnecessary duplication is reduced, and data consistency is maintained.

First Normal Form (1NF)

The database follows 1NF by ensuring that each table has a primary key and that each field contains a single, atomic value. Repeating groups and lists of values are avoided within a single field.

Example of an unnormalized design:

Customer_ID	Customer_Name	Phone_Numbers
001	Juan	0912, 0913, 0914

Normalized design:

Customer_ID	Customer_Name	Contact_Number
001	Juan	0912

Second Normal Form (2NF)

The database is designed to satisfy 2NF by building on 1NF and keeping attributes dependent on the record's primary key. Customer information is kept in CUSTOMERS, computer information in COMPUTERS, technician information in TECHNICIANS, and service information in REPAIR_SERVICES instead of repeatedly storing the same information in transaction records.

Third Normal Form (3NF)

The database is designed toward 3NF by keeping information that describes a specific entity in its appropriate table. For example, technician details are stored in TECHNICIANS rather than repeated in every REPAIR_REQUEST, while service details are stored in REPAIR_SERVICES. Foreign keys are then used to connect related records.

Normalization of the Project Tables

Table	Normalization Purpose
CUSTOMERS	Stores customer information separately to avoid repeating customer details.
COMPUTERS	Stores computer information and links each computer to its customer.

APPOINTMENTS	Stores appointment information separately from customer and computer details.
REPAIR_REQUESTS	Stores the main repair transaction and references the related computer, technician, and service.
TECHNICIANS	Stores technician information separately so it does not need to be repeated for every repair.
REPAIR_SERVICES	Stores service information separately so service details are not duplicated in repair records.
PAYMENTS	Stores payment information separately and links each payment to the related repair request.

Final Normalization Statement

The Computer Repair Appointment and Service Management System is designed with a normalized relational structure. The database separates customer, computer, appointment, repair request, technician, repair service, and payment information into individual related tables. Each table uses a primary key to identify records, while foreign keys establish relationships between related tables. This structure reduces unnecessary data duplication and helps maintain data consistency.